

A Question Answering System for Aerospace Large Language Models Based on Knowledge Graph and RAG Collaboration

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Abstract

At present, the data in the aerospace science and technology is massive, heterogeneous, distributed and discrete, and lacks unified management, which seriously affects the quality and efficiency of question retrieval. To solve this problem, a construction method of question answering system for aerospace large language models based on knowledge graph and retrieval enhanced generation (RAG) collaboration is proposed. Firstly, the aerospace science and technology information representation model is constructed, and then the aerospace science and technology information resource database is constructed based on the model. On this basis, the large language model in the aerospace field is trained by using the efficient fine-tuning technology; Create knowledge graph and vector database in the neo4j, and use entity attributes and vector indexes to integrate them; Finally, the langchain framework is used to link the knowledge graph, RAG and the large language model of the aerospace field to develop and construct the aerospace intelligent question answering system. The experimental results show that the system covers a wide range of knowledge in the aerospace science and technology field, and can effectively improve the accuracy of the answer and reduce the hallucination rate.

CCS Concepts

• Information systems; • Information retrieval; • Retrieval models and ranking; • Language models;

Keywords

intelligent question answering, large language model (LLM), knowledge graph, retrieval augmented generation (RAG), the aerospace science and technology

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1 Introduction

With the rapid development of aerospace science and technology, a large number of heterogeneous aerospace science and technology data are generated. In the process of obtaining and applying these data, we often encounter many difficulties [1], mainly including the following aspects: (1) Aerospace science and technology data types span a large range, covering articles, figures and scientific and technological products. These data are stored on different platforms and have different formats, so it is difficult to integrate and share the data; (2) The knowledge in the field of aerospace is professional, involving complex physics, engineering and computer science theories, which makes it difficult for ordinary people to master and use; (3) At present, the existing aerospace science and technology data retrieval system mainly adopts the way of keyword search, which has a weak understanding of semantics and can not provide high-precision targeted search, so it is difficult to meet the actual needs of scientific research and engineering practice; (4) Traditional data analysis methods [2] can not effectively analyze and mine data; (5) At present, aerospace knowledge is continuously and rapidly updated. For example, as soon as new concepts such as "commercial aerospace" and "low altitude economy" are put forward, a large number of relevant knowledge are constantly emerging. However, most of the existing knowledge management systems [3] do not have dynamic update mechanism and lack the ability of real-time update, so it is difficult to reflect the latest research progress and engineering application results in time for these systems.

In recent years, with the rise of large language model (LLM), its integration with intelligent question answering system [4] has gradually become a hot research direction [5-6]. However, there are a large number of complex professional questions in the field of aerospace science and technology. When the LLM answers these professional questions, it often produces false, wrong or even harmful information, which is prone to hallucinations [7], which makes people start to worry about whether its answer is reliable. Recently, the combination of knowledge graph [8], retrieval enhanced generation (RAG) [9] and LLM has gradually become a new problem-solving idea, which shows great potential in the field of intelligent question answering. According to the needs of aerospace science and technology, this paper proposes and designs a question answering system for aerospace large language models based on

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knowledge graph and RAG collaboration, and has been effectively applied.

2 Related technology research

2.1 Fine tuning of large language model

Fine tuning technology^[10-11] can significantly improve the performance of the model on specific tasks. Taking the mainstream pre training models such as BERT^[12], RoBERTa^[13], T5^[14] and GPT^[15] as examples, these basic models based on massive general data training can quickly adapt to the new knowledge system after fine-tuning the annotation data in specific fields. However, due to the large data demand, this technology is not suitable for simple application scenarios.

2.2 Prompt learning

Prompt learning^[16] is a new model guidance technology, which effectively stimulates the potential ability of the pre-training model with the help of natural language instructions. Its core mechanism is to embed carefully designed text cues in the original input, which can effectively guide the model to produce output that meets the user's expectations. This method is particularly useful for task scenarios where the model has mastered the relevant knowledge base. It can maintain the flexibility of the model and significantly improve the efficiency of task execution. However, the design of the prompt template depends on human experience. There is no clear standard for prompt optimization. The debugging cycle is usually very long and the effect is difficult to determine.

2.3 Knowledge Graph

Knowledge graph (KG)^[17] is a semantic knowledge base that uses the pattern of graph to organize knowledge, uses a node to represent an object, and uses the lines between nodes to represent the relationships between objects. Its most basic unit is triplet^[18] (entity - relationship - entity or entity - attribute).

Knowledge graph can enhance the LLM in the direction of external knowledge^[19-21], and then improve the transparency and interpretability of the system, making the system suitable for scenarios that need to deal with structured knowledge and complex relationships. However, due to the limitations of the query method of knowledge graph, once there are errors in the process of querying entities or relationships, the query results will not be found or the wrong retrieval results will be returned, or even errors will be reported. In addition, it is difficult to construct the knowledge graph, and the information in the real world is constantly changing, which makes it difficult to ensure the timeliness of knowledge.

2.4 Retrieval Augmented Generation

Retrieval augmented generation (RAG) combines the two modules of information retrieval and text generation, and has significant advantages in natural language processing tasks. Based on the questions raised by users, this technology retrieves relevant information fragments from the knowledge base, submits the results and initial questions to the LLM, and improves the reliability and accuracy of the content generated by the LLM with the help of the external knowledge base^[22-23], which is suitable for the scene

of real-time knowledge update. However, its enhancement effect largely depends on the retrieval module. If the retrieved results are not highly relevant to user problems, some irrelevant information will be introduced, which will affect the quality of generated content.

2.5 LLM + RAG - KG

LLM + RAG - KG^[24] is a technical framework that creatively integrates the three major technical modules of large language model, retrieval augmented generation and knowledge graph to build a synergy mechanism with complementary advantages. In this architecture, the structured knowledge system of knowledge graph brings two benefits to the system: first, it improves the accuracy of semantic parsing of rag module; Secondly, it provides a reliable structural basis for the model reasoning process. This organic combination of structured knowledge and neural network model not only retains the powerful generation ability of LLM, but also greatly improves the accuracy and interpretability of the answer through the constraint mechanism of knowledge graph, showing unique technical advantages in the field of intelligent question answering.

3 Research on the expression paradigm of Aerospace Science and technology data

This research deeply analyzes the representation methods of aerospace science and technology data, and constructs the aerospace science and technology information representation model on this basis. The model determines the categories, attributes and relationships of data, which facilitates the unified management of data. After analyzing the influencing factors of knowledge retrieval and application in aerospace field, this study selected the following six main data categories:

- Article: Papers, patents and scientific and technological achievements are the basis of aerospace technology innovation. Such data can support technology retrieval, frontier research tracking and scientific knowledge sorting.
- Event: Event in the aerospace field, such as news and economic information, is updated faster than article, and can better ensure the timeliness of the intelligent question answering system.
- Spacecraft: Spacecraft equipment, such as deep space detectors, satellites and rockets, is the core content of aerospace science and technology. Such data can provide corresponding data basis for functional modules such as technical analysis.
- Institution: The development of aerospace science and technology cannot be separated from relevant institutions, such as the space administration, scientific research institutes, aerospace science and technology enterprises, etc. With the help of such data, users can clearly understand the functional scope, cooperation and main achievements of each institution.
- Figure: Mainly including experts, scholars and astronauts in the aerospace field. Such data is very important for academic research, talent tracking and reviewing major historical events.

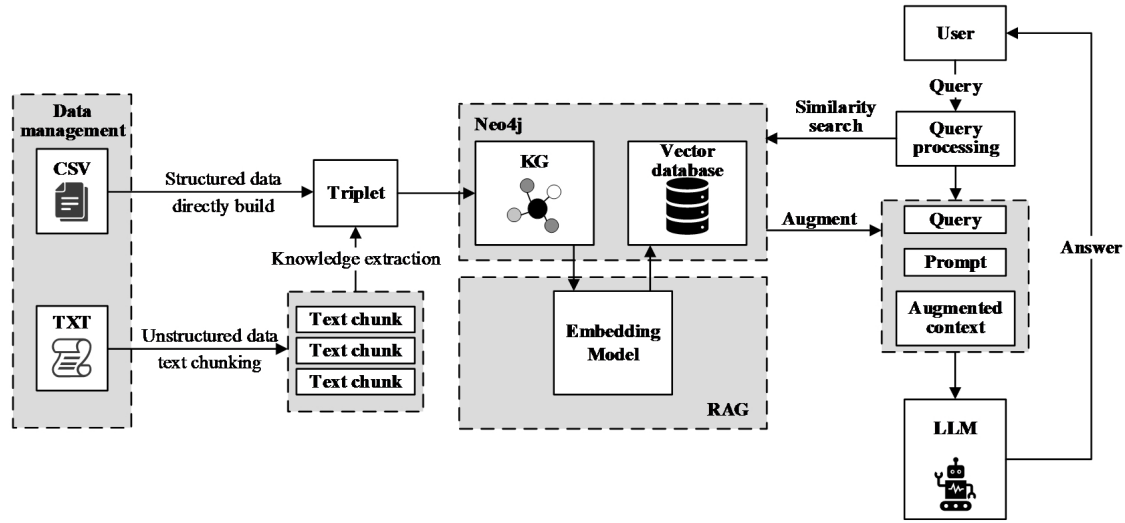


Figure 1: Overall architecture.

- Regulations and guides (RGs): Space activities are strictly restricted by international and domestic laws and regulations. Adding such data can facilitate users to retrieve relevant legal provisions and understand the specification requirements.

On this basis, according to the characteristics of multi-source and heterogeneous aerospace science and technology data, this study designed the corresponding data integration framework for the six data categories. These frameworks are used to effectively integrate specific types of data, thus promoting the integration of similar data.

Relying on the above model and framework, this study built a high-quality aerospace science and technology information resource database, which laid the foundation for the follow-up work of knowledge management, intelligent retrieval and question answering system.

4 System framework

4.1 Overall architecture

The overall architecture of the question answering system for aerospace large language models based on knowledge graph and RAG collaboration is shown in “Figure 1”.

The architecture mainly includes four parts: data management, knowledge graph, the collaboration between RAG and knowledge graph, and large language model call.

4.1.1 Data management. After studying the expression paradigm of aerospace science and technology data, this study identifies six knowledge categories of aerospace science and technology information resource database: article, event, spacecraft equipment, institution, figure, regulations and guides. And it is planned to gradually determine the specific data sets under each major category in the subsequent data collection and data pre-processing.

Data collection: there are more than 30 sources of data, including: a) Article: CNKI, Google academic, IEEE explore, etc; b) Event:

China News network, financial headlines, etc; c) Spacecraft: National Space Administration, Space Factory, China Manned Space Engineering network, China Lunar Exploration and Deep Space Exploration network, etc; d) Institution: Qizhidao, National Space Administration; e) Figure: Baidu Encyclopedia, CNKI; f) RGs: China Law retrieval system, the website of the Ministry of science and technology of China. The corpus data is collected from the above data sources by the method of combining manual collection and automatic collection.

Data pre-processing: import the collected original data into the data set for pre-processing operations such as cleaning, conversion, filtering, de-duplication and de-privacy to improve the data quality.

Data annotation: after pre-processing, the data sets are grouped and annotated by combining manual and automatic annotation. The questions are extracted from the data sets and the answers are annotated. All data labels are manually reviewed and revised. For some data that are difficult to label, labels can be used to facilitate subsequent targeted labeling or review. High quality aerospace science and technology knowledge Q&A pairs of labeled data are formed through experts in the field.

In the process of improving the data set, this study gradually build a professional aerospace science and technology information resource database, as shown in “Figure 2”. Finally, 12847 pieces of high-quality data are integrated into the resource database, 70% of which are used as training sets for parameter fine-tuning of the model in the field, and the remaining 30% of which are used as validation sets to evaluate the performance of the model.

In the design of data storage scheme, for highly structured data, CSV format is used to ensure the integrity of data fields, which is also convenient for the subsequent construction of knowledge graph; For unstructured text data, TXT format is selected for storage. This flexible storage method provides convenience for subsequent text block processing.

4.1.2 Knowledge graph. Neo4j is selected as the graph database.

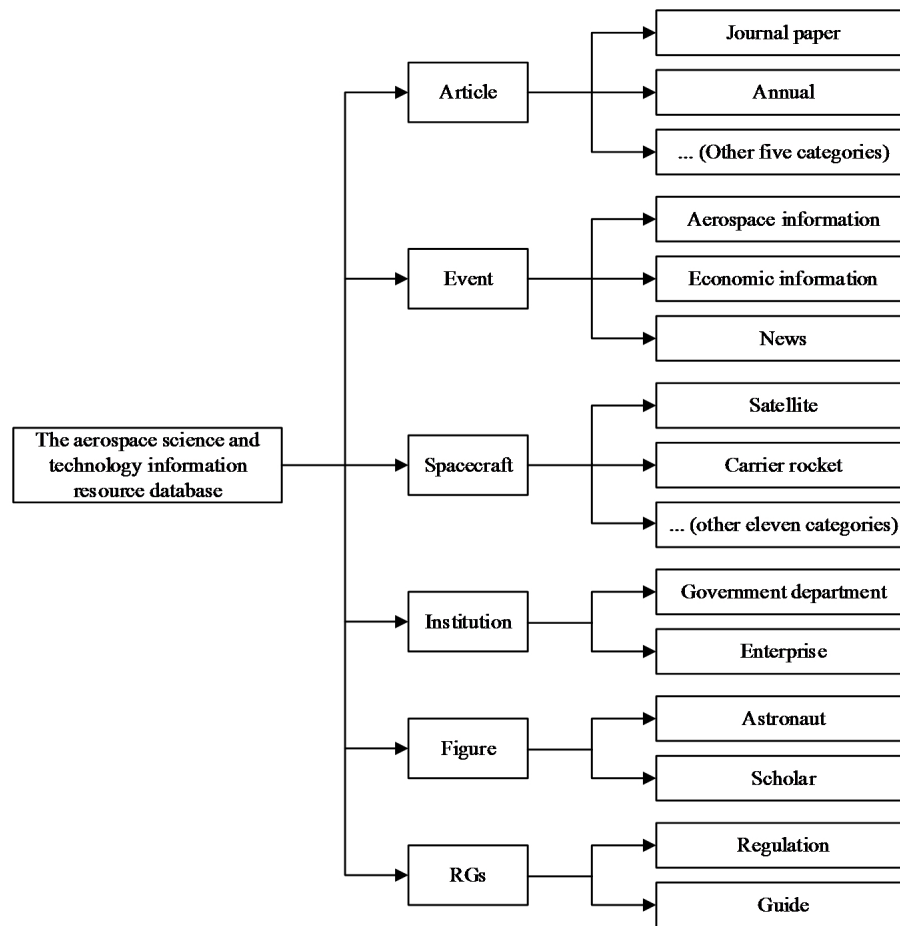


Figure 2: The aerospace science and technology information resource database.

For structured data, after the triples are created according to the fields, they can be directly imported into neo4j. For unstructured text, it is necessary to divide the original text into multiple text chunks first, create corresponding nodes for each chunk in the knowledge graph, and store the text information in the "text" attribute of the corresponding node. Then we extract knowledge with the help of LLM to obtain the triplet, and then build a knowledge graph on this basis. Part of the knowledge graph is shown in "Figure 3".

4.1.3 The collaboration between RAG and knowledge graph. Select the "gte-large-zh" in modelscope as the embedding model.

Complete the creation of all vectors with the embedding model, and store the created vectors as attributes in the corresponding nodes or relationships. After specifying a unique index name, create an index for the vector, build a complete vector database in neo4j, and integrate the vector database with the knowledge graph. At this point, the effective collaboration between RAG and knowledge graph can be realized: when the user inputs a query request, the system first uses the vector index to retrieve the entities and relationship fragments most related to the query semantics in the knowledge graph as supplementary information for retrieval, and

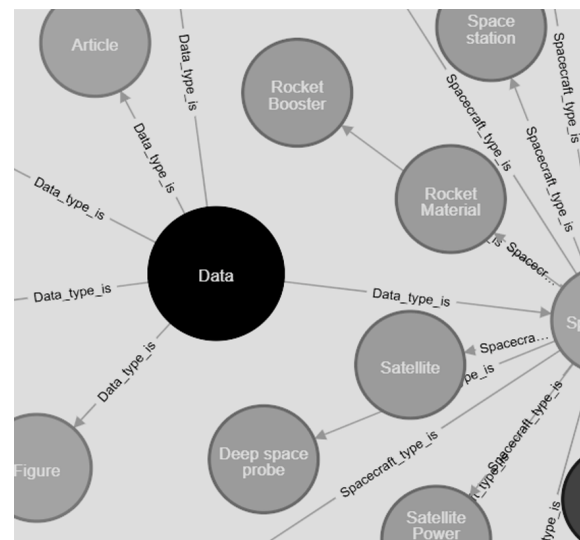


Figure 3: The partial display of knowledge graph.

then inputs these knowledge fragments together with the original question into the LLM for generative answer, so as to take into account the accuracy of knowledge and the flexibility of generation, and improve the accuracy and interpretability of the question answering system.

4.1.4 Large language model call. “Qwen-7b-chat” is selected as the basic pre-training model, and it is trained as an exclusive model in the aerospace field using Lora fine-tuning technology.

After receiving the question sent by the user, the searcher will use the cosine similarity algorithm in neo4j for similarity search to obtain the sub graph containing the relevant information of the question. After the query of qualified sub graphs, the system will provide the queried sub graphs (structural information) together with questions and prompts to the LLM as input. The LLM generates the most appropriate answers to feed back to users by analyzing the semantic and structured information input.

4.2 The key technologies

The main flow chart of the question answering system for aerospace large language models based on knowledge graph and RAG collaboration is shown in “Figure 4”.

The system adopts six key technologies, including the efficient fine-tuning technology based on Lora, the automatic construction technology of knowledge graph, the dynamic update technology of knowledge graph, the vector retrieval technology based on the combination of knowledge graph and RAG, the simplification technology of complex problems and the technology supporting multiple rounds of dialogue.

4.2.1 The efficient fine-tuning technology based on Lora. Lora is an efficient parameter tuning technology. By introducing a low rank decomposition matrix as an adapter, it can achieve significant performance improvement by adjusting a small number of key parameters in the model. Based on this feature, Lora has become an ideal choice for this research to achieve model domain adaptation. By organically combining the general pre-training model with the professional knowledge of aerospace science and technology, this research successfully constructs a special model for aerospace field with low computational cost on the premise of ensuring the performance of the model.

4.2.2 The automatic construction technology of KG. Knowledge graph is one of the core of question answering system, which plays the role of graph database and vector database at the same time. Therefore, it is very important to build a high-quality knowledge graph. Furthermore, triplets are the key factors affecting the quality of knowledge graph.

For structured data, it is relatively simple to directly create effective triplets based on fields.

For unstructured text, we need to design a knowledge extraction method that can obtain effective triplets. The system first blocks the long text, and stores the chunked text chunks as entities in the knowledge graph. The maximum length of each chunk is set to 100 tokens, and the repeatable length is set to 20 tokens. The Chinese and English comma, period and other separators are used as the blocking conditions. After that, the prompt words of the cases containing the extracted triplets are constructed, and then

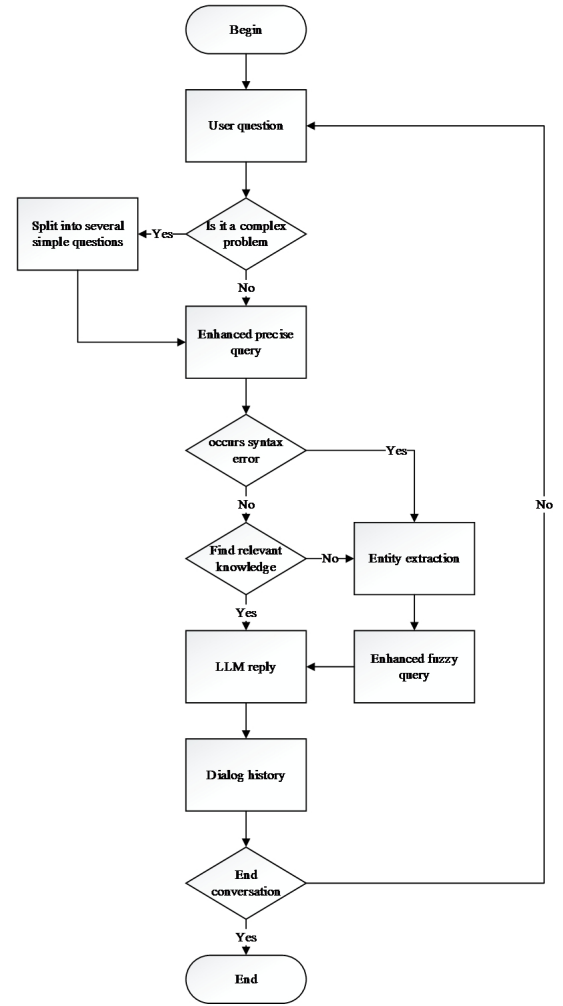


Figure 4: The flow chart of the question answering system.

the prompt words are sent to the LLM. With the help of the LLM, the knowledge extraction is completed, and the quality of the final triplets is further improved.

4.2.3 The dynamic update technology of knowledge graph. In this study, an intelligent dynamic updating mechanism of knowledge graph is designed to ensure the timeliness and accuracy of knowledge base through multi-stage collaborative processing. The mechanism uses the change detection technology based on time stamp, combined with vector space matching and cosine similarity algorithm to achieve the accurate alignment of new entities. In terms of update strategy, the system implements a dual guarantee mechanism: on the one hand, it maintains data consistency through the coverage update algorithm, and on the other hand, it uses the elimination mechanism based on the time decay function to automatically clean up low value data.

4.2.4 The vector retrieval technology based on the combination of KG and RAG. The vector retrieval technology designed in this study is an important guarantee for the question answering system to

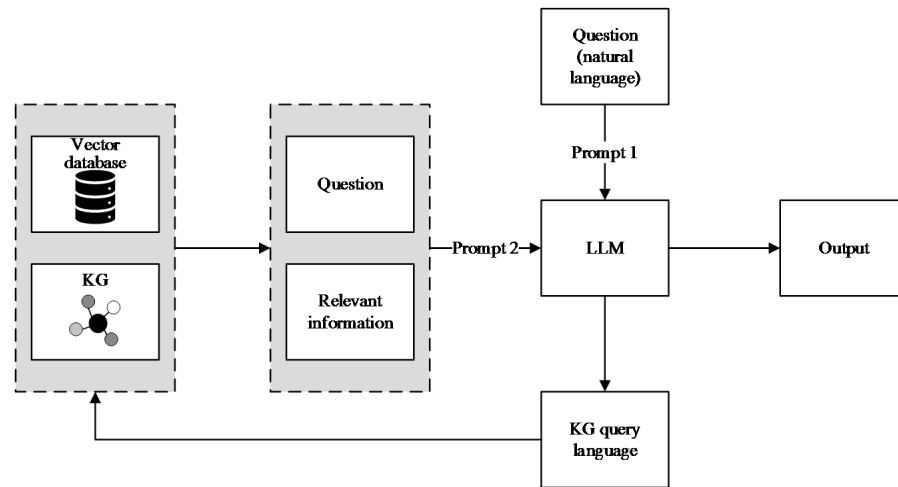


Figure 5: The precise query.

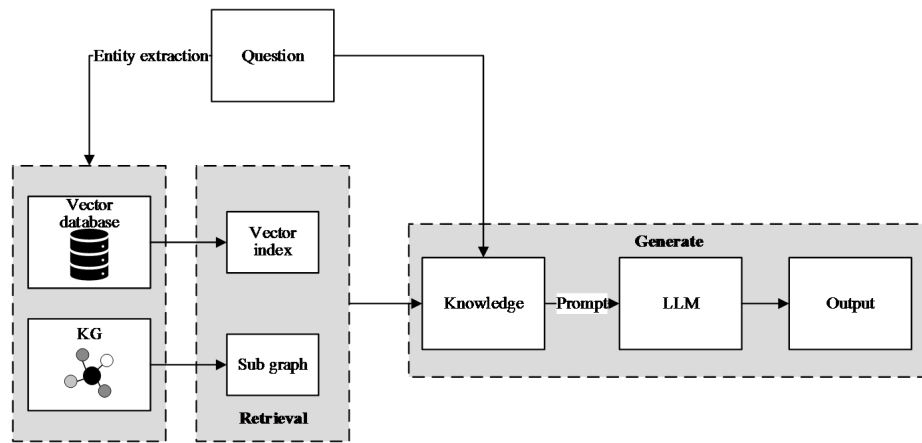


Figure 6: The fuzzy query.

generate high-quality answers. This technology uses two retrieval methods.

The first method is precision query, and its architecture is shown in “Figure 5”. Precise query uses LLM to transform the user’s questions from natural language to the query language of graph database, and then retrieve the qualified sub graph information in neo4j. This method uses the query language, so its accuracy is very high, but its retrieval effect is largely determined by the quality of the query language. If the LLM generates the wrong query language, it will not be able to query any effective information, and even error may occur due to syntax errors. This study uses a large number of cases that convert natural language into query language (most of them belong to the aerospace field, and a few belong to other fields) to fine tune the pre-training model, and adds corresponding prompts in the prompt words, which effectively improves the quality of query language and forms an enhanced version of accurate query with higher fault tolerance.

The second retrieval method is fuzzy query, and its architecture is shown in “Figure 6”. Fuzzy query uses vector similarity to retrieve the qualified sub graph information from the knowledge graph. This retrieval method can not achieve the high accuracy of precise query, but its fault tolerance rate is very high and its application scope is wider. Through experiments, this study found the similarity search parameters applicable to all vector types (the knowledge quantity with the highest similarity score returned - K, and the minimum similarity score limit of the returned knowledge - score_threshold) and summarized them. When it is necessary to use fuzzy query to retrieve a specific vector type, the accuracy of the query is effectively improved with the parameters read from the summary, forming an enhanced version of fuzzy query. Enhanced fuzzy query has higher query accuracy.

After comprehensively considering the advantages and disadvantages of the two retrieval methods, the research finally adopted the strategy of mutual cooperation between the two, and put the enhanced version of accurate query in a higher priority position,

which not only ensured the accuracy of the question and answer system, but also ensured the stability.

4.2.5 The simplification technology of complex problems. When the initial problem raised by the user is relatively complex and needs to be solved step by step, if the entire initial problem is retrieved directly, the multiple questions contained in it will interfere with each other, resulting in poor retrieval effect. Therefore, before retrieval, the research uses the LLM for semantic recognition. If the problem is indeed relatively complex, it will segment the complex problem, and supplement the missing semantics according to the background, and finally get multiple simple sub problems with complete semantics and clear meaning. At the same time, the LLM can also identify the entities that need to be retrieved for each simple sub problem, and then it can be retrieved directly for the entity type, while ensuring the accuracy and efficiency of the query. Finally, after finding the most appropriate answers to all the simple sub questions, the LLM integrates the answers and returns them to the user.

4.2.6 The technology supporting multiple rounds of dialogue. Save the question and answer record between the user and the LLM in the session storage, and then take out the question and answer record and add it to the prompt word when continuing the question and answer. By sending the prompt word to the LLM, the support for multiple rounds of dialogue question and answer is realized, and the accuracy and stability of the whole question and answer system are improved.

5 Application examples

5.1 Application

After the research on the acquisition principle and expression paradigm of aerospace science and technology data, the research identified six knowledge categories of aerospace science and technology information resource database, and formed and continuously improved the data sets under each category in the process of data collection, data pre-processing and data annotation. Finally, the aerospace science and technology information resource database was established, with 12847 pieces of data, including 9000 pieces of training data, accounting for 70%, and 3847 pieces of validation data, accounting for 30%.

“Qwen-7b-chat” is selected as the basic pre-training large language model, and Lora fine-tuning technology is used to complete the fine-tuning training of LLM in the field of aerospace science and technology based on the marked data.

Select neo4j as the graph database. For the structured data stored in the CSV file, analyze its fields to determine the corresponding triplets and store them in neo4j. For unstructured text stored in TXT file, after setting the relevant parameters and segmentation conditions of text segmentation, the original text is segmented and the segmented text chunk is stored as an entity in neo4j. After that, the prompt words containing the extracted triplet cases are constructed, and then the prompt words are sent to the LLM. With the help of the LLM, the knowledge extraction of all chunks is completed, and the high-quality triplet is obtained and stored in neo4j. In the final knowledge graph, the “source” attribute of the

> **Entering new GraphCypherQAchain chain ...**

Generated Cypher :

MATCH (m : Rocket_Material)

RETURN m.value AS NameOfSpacecraft LIMIT 3

Full Context:

```
[
  { 'NameOfSpacecraft' : ' Silicon carbide fiber ' },
  { 'NameOfSpacecraft' : ' Alumina ceramic ' },
  { 'NameOfSpacecraft' : ' Carbon composite ' }
]
```

> **Finished chain .**

Figure 7: Process example I.

entity indicates the source of the data, and the “timestamp” attribute stores the time stamp of the entity change.

Select the “gte-large-zh” model as the embedding model. In the constructed knowledge graph, the model is used to create vectors for the specified objects. After the vector is created, the index is created according to its category. Finally, vector database and knowledge graph are integrated into neo4j. In case of subsequent data changes, the dynamic updating method of the knowledge graph described in section IV.B.3) can be used to maintain the real-time performance of the knowledge graph and the consistency with the vector database.

The question answering system uses the langchain framework to integrate the vector retrieval method described in section IV.B.4), the complex problem decomposition and simplification method described in section IV.B.5), and the method supporting multiple rounds of dialogue described in section IV.B.6).

After the user sends a request, the system will automatically give the most appropriate answer in combination with the knowledge graph. Specific examples are as follows:

The user sent a request: “Give the names of three rocket materials”. The question answering system first retrieved the relevant information in the knowledge graph, and the retrieval process is shown in “Figure 7”. Then, the answer is given in combination with the user’s request, and the result is shown in “Figure 8”.

The user sent a request again: “Give the types and applicable temperature ranges of the above three rocket materials respectively”. The question and answer system split and simplified the complex problem. Through the historical dialogue, it was able to understand the specific materials referred to by “the above three rocket materials”, retrieve the relevant knowledge of each simple sub problem in the knowledge graph. The retrieval process is shown in “Figure 9”. Finally, the answers are summarized and returned, as shown in “Figure 10”.

5.2 Impact assessment

The quality of the answers provided by the question answering system and the severity of hallucination problems were quantitatively evaluated through two indicators of similarity and hallucination

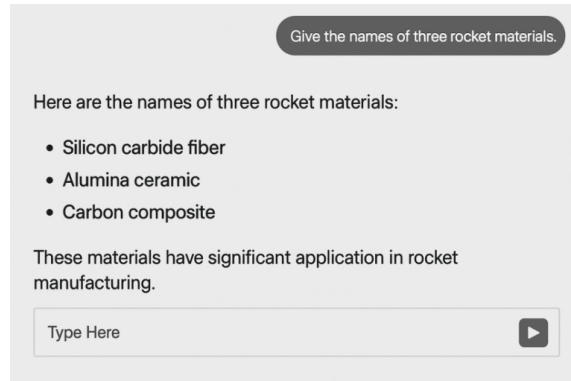


Figure 8: Result example I.

rate, and the different performance before and after fine-tuning and after using LLM + RAG - KG technology were analyzed for different target scenarios.

5.2.1 Similarity analysis. Measure the similarity between the generated answer and the corresponding marked answer of the question answering system, and conduct it on the verification set. The similarity calculation method is as follows: the generated answer and the marked answer are cleaned, converted and other pre-processing operations respectively, and then the “gte-large-zh” model is used to generate the text embedding vectors of the two kinds of answers respectively, and then the cosine similarity between the two embedding vectors is calculated. Finally, the results are converted into percentage form. The closer the value is to 100%, the closer the generated answer is to the marked answer. “Table 1” shows the results of each target scene before and after fine-tuning and after using LLM + RAG - KG technology.

As can be seen from “Table 1”, the Overall similarity has been greatly improved after fine-tuning, with an increase of 17.9 percentage points. Especially in the target scenarios such as Article, Event and Spacecraft, the improvement effect is more significant because of the large number of labeled data and certain diversity. After further using LLM + RAG - KG technology, the Overall similarity increased by 4.3 percentage points again, and the largest increase was in Spacecraft, Institution and Figure, especially Spacecraft, which increased by 7.4 percentage points. This is mainly because most of the knowledge in such scenes comes from structured data. Structured data can form high-quality triplets stably according to

its fields, and the sub graphs constructed based on such triplets are more systematic, standardized and clear. However, the knowledge in the Article, Event and RGs is more from unstructured text, and triplets cannot be obtained directly. Triplets can be obtained only after a series of operations such as text segmentation and knowledge extraction. In this process, there are multiple factors that affect the quality of triplets. The integrity and accuracy of sub graphs constructed from such uneven triplets must be lower than those of the first three categories, so the effect of improving the similarity of the latter three categories of scenarios is relatively general.

5.2.2 Hallucination rate analysis. The hallucination rate of the question answering system refers to the proportion of the answers with hallucinations in the total number of questions. It is an important indicator to measure the authenticity of the answers and the severity of hallucinations in LLM. In order to determine whether the answers generated by the model have appeared illusion, three experts in the field of aerospace were specially invited to make a simple vote, and the option with more votes is regarded as the final decision result. The results show that LLM + RAG - KG technology can effectively reduce the hallucination rate of question answering system, and the overall hallucination rate has decreased by 8.7%, from 16.3% to 7.6%. In addition, the hallucination rate of the question answering system can be maintained at a low level by using the dynamic update technology of knowledge graph to update knowledge in real time.

6 Conclusion

This research provides a complete solution for data management and intelligent question answering in the field of aerospace science and technology. By constructing a question answering system for LLM based on KG and RAG collaboration, three application values are realized: at the engineering practice level, the system can provide intelligent support for the whole life cycle of space missions, including professional knowledge retrieval in the scheme design stage, real-time decision-making assistance in the task execution process, and knowledge traceability in the later evaluation; In the dimension of knowledge dissemination, the system has broken the professional barriers, which can not only be used as a teaching aid for aerospace educators, but also carry out accurate science popularization for the public; In terms of technology migration, the proposed methodology has significant cross domain applicability, and its core technology framework can be quickly adapted to intelligent system development in other professional fields.

Table 1: Similarity analysis %.

Target scenario	Before fine-tuning	After fine-tuning	LLM + RAG - KG
Overall	59.7	77.6	81.9
Article	57.2	76.1	80.0
Event	53.4	74.3	76.4
Spacecraft	57.7	77.4	84.8
Institution	63.3	79.2	83.6
Figure	66.7	81.9	86.3
RGs	59.8	76.6	80.1

> Entering new GraphCypherQAchain chain ...

Generated Cypher :

MATCH (m : Rocket_Material)

WHERE m.value IN ["Silicon carbide fiber " , " Alumina ceramic " , " Carbon composite "]
RETURN m.value AS NameOfSpacecraft , m.Type AS Type

Full Context:

```
[
  {'NameOfSpacecraft': 'Silicon carbide fiber ', 'Type': 'Insulation material '},
  {'NameOfSpacecraft': 'Alumina ceramic ', 'Type': 'Ceramic material '},
  {'NameOfSpacecraft': 'Carbon composite ', 'Type': 'Composite material '}
]
```

> Finished chain .

(a) Sub question I

> Entering new GraphCypherQAchain chain ...

Generated Cypher :

MATCH (m : Rocket_Material)

WHERE m.value IN ["Silicon carbide fiber " , " Alumina ceramic " , " Carbon composite "]
RETURN m.value AS NameOfSpacecraft , m.Temperature_range AS TemperatureRange

Full Context:

```
[
  {'NameOfSpacecraft': 'Silicon carbide fiber ', 'TemperatureRange': '1200°C - 1600°C '},
  {'NameOfSpacecraft': 'Alumina ceramic ', 'TemperatureRange': '1400°C - 1800°C '},
  {'NameOfSpacecraft': 'Carbon composite ', 'TemperatureRange': '3000°C (short time)'}
]
```

> Finished chain .

(b) Sub question II

Figure 9: Process example II.

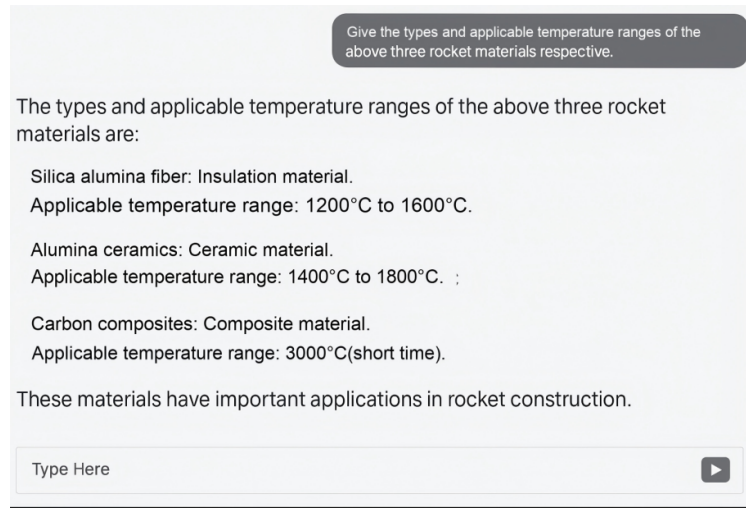


Figure 10: Result example II.

Future research can further optimize the dynamic updating mechanism of knowledge graph and continuously enhance the adaptability of the system in the rapidly developing aerospace science and technology field. At the same time, considering the general advantages of the system in heterogeneous data integration, professional knowledge modeling and intelligent question answering, we can continue to explore its application potential in complex knowledge intensive fields such as the health care, the legal consulting and the education services in the future.

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